

Mathematical Performance Analysis (Tailored to Our Pipeline)

A. Little's Law for our streaming ingestion

We model the streaming path (publisher → Pub/Sub subscription → consumer) as a queueing system.

Let:

- λ = average arrival rate of events (events/second)
- W = average end-to-end time an event spends in the system (seconds)
- L = average number of events present in the system (backlog in subscription / in-flight)

Little's Law: $L = \lambda \times W$

How to interpret for our demo:

- If λ increases (we publish faster) but consumer speed stays constant, W and L increase.
- If we improve consumer throughput, W and L drop for the same λ .

B. Service rate and stability condition

Let μ be the consumer service rate (events/second). For stable processing, we need:

$$\rho = \lambda / \mu < 1$$

ρ is utilization. If ρ approaches 1, queue delays grow quickly. If $\rho > 1$, backlog grows without bound.

In our pipeline, μ depends on:

- message parsing cost
- network latency
- downstream write/processing steps (if enabled)

C. Practical measurement using logs

We can estimate λ and μ from timestamps in logs:

- $\lambda \approx (\text{number_of_published_messages}) / (\text{publish_time_window})$
- $\mu \approx (\text{number_of_consumed_messages}) / (\text{consume_time_window})$

Then estimate W from:

$$W \approx (\text{mean consume_timestamp} - \text{publish_timestamp})$$

Finally compute L using Little's Law:

$$L \approx \lambda \times W$$

This gives a quantitative explanation of observed backlog and delays when we speed up the publisher.

D. Batch windowing and aggregation latency

For the daily aggregation job (Airflow + BigQuery), we treat each run as a batch window.

Let:

- T = batch window size (e.g., 1 day)
- D = DAG run duration (seconds)

A simple SLA-style metric is batch freshness latency:

Freshness $\approx$ D (how long after the window ends until aggregates are available)

If D grows, analytics and ML retraining become stale. Our goal is to keep D low by using SQL aggregations in BigQuery.

E. ML error metric connection

Our regression model minimizes Mean Squared Error (MSE):

$$\text{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$$

In BigQuery ML evaluation we also observe RMSE and MAE. Lower values indicate predictions closer to actual revenue. This ties back to data quality and feature stability from the upstream pipeline.